

Retrieval-Augmented Generation (RAG)

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Large Language Models are confident, but isolated.

Knowledge Cutoff



Model knowledge is frozen at the end of its training data.

Hallucinations



When lacking data, the AI attempts to imagine and present false information as truth.

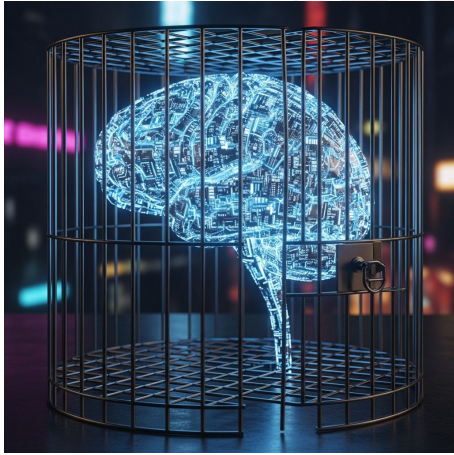
Lack of Privacy



Public models have zero visibility into your organization's secure, internal data.

The open-book test for Artificial Intelligence.

Retrieval-Augmented Generation (RAG) makes AI smarter by letting it **search** for real information from **trusted sources** before it answers you.



Without RAG:

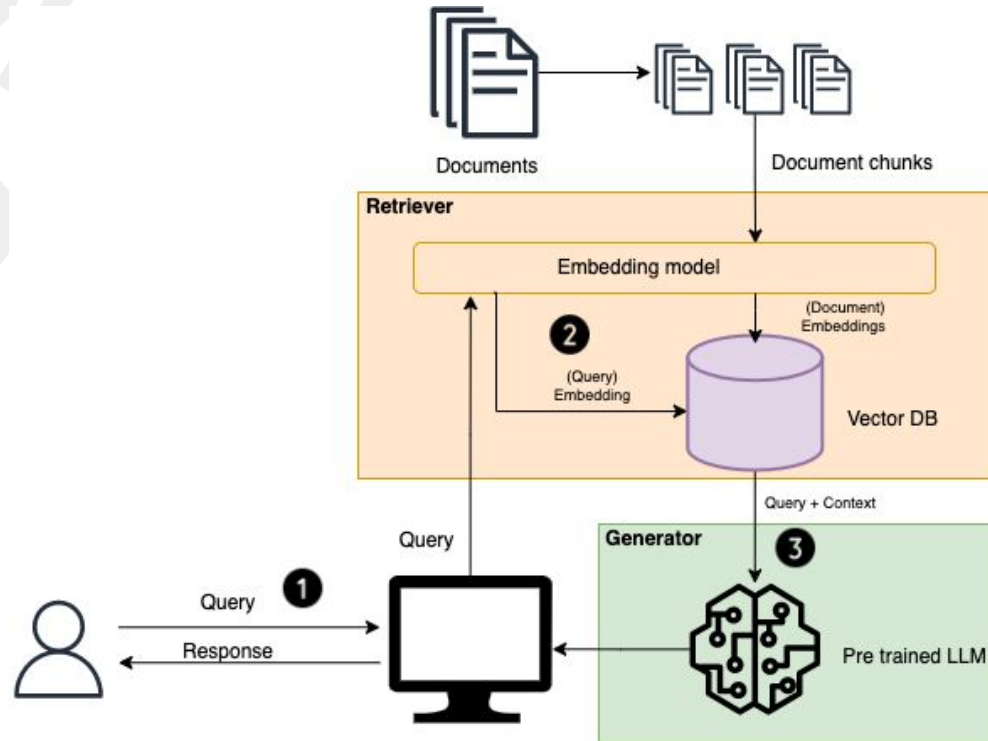
AI only uses its **memory**. If it doesn't know the answer, it might **"hallucinate"** (make things up).



With RAG:

AI **checks the facts** from your provided files to give **accurate** and **up-to-date** answers.

Retrieval-Augmented Generation (RAG)



Four enterprise pillars of RAG architecture.



Cost-Effectiveness

It **minimizes** expenses by avoiding the need for expensive model fine-tuning. You only need to update your documents.

Current & Fresh

The system **maintains** up-to-date information by connecting directly to live databases and recent policy updates.



<https://www.dataversity.net/articles/types-of-database-management-systems/>



Trust & Transparency

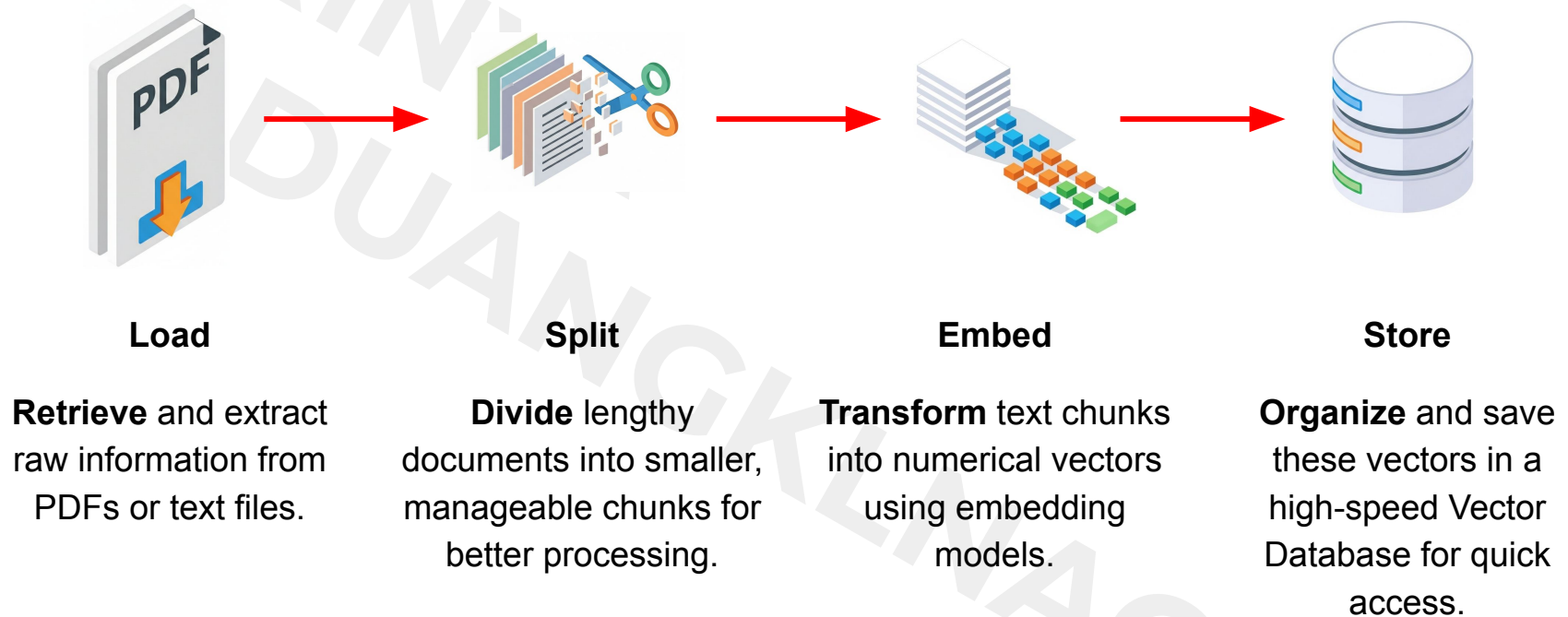
It **ensures** reliability by requiring the AI to cite specific sources for every answer.

Control

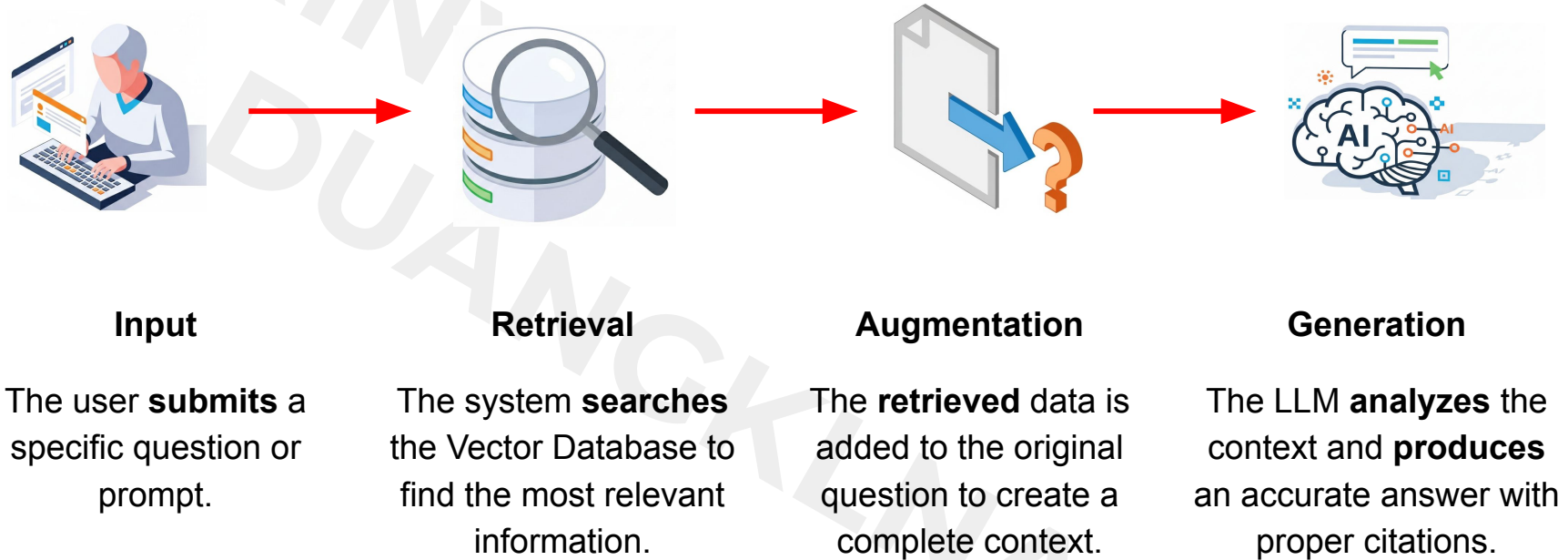
It **restricts** access to sensitive information, allowing developers to manage what the AI is permitted to see.



Phase 1: The Indexing Pipeline (Data Preparation)



Phase 2: Processing the User Query



Elevating retrieval with Semantic Search.

The Problem:

Standard keyword searches often **fail** to find information in complex enterprise documents because they only look for exact word matches.

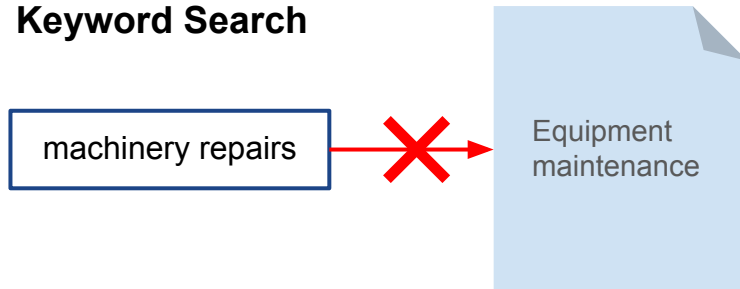
The Solution:

Semantic search **identifies** the user's **intent** and the **context** of the query, rather than just matching individual words.

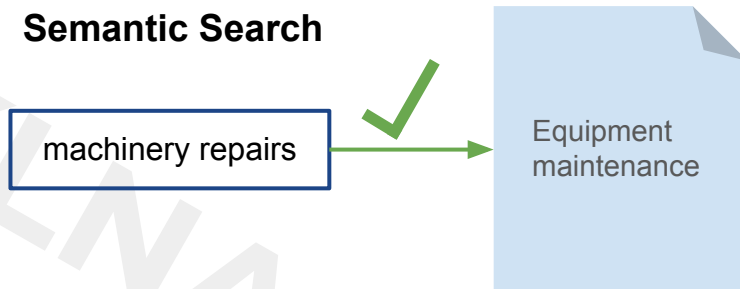
Example:

If a user asks about "**machinery repairs**," the system can **retrieve** relevant data even if the document uses different terms like "**equipment maintenance expenditures**."

Keyword Search



Semantic Search



Advanced optimization: Maximum Marginal Relevance (MMR).

The Problem:

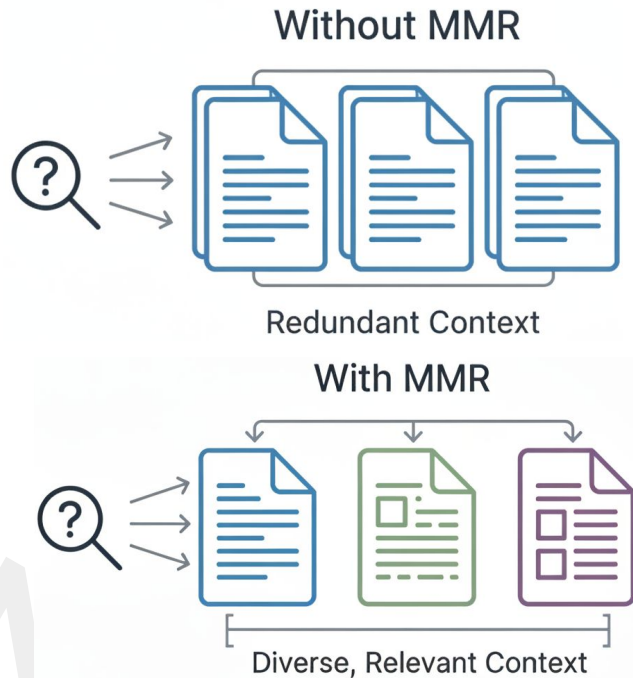
Standard semantic search often **returns** multiple pieces of information that are too similar, leading to **redundant** and repetitive results.

The Fix:

MMR is an algorithm that **balances Accuracy** (how relevant the information is) with **Diversity** (ensuring the results cover different perspectives or facts).

Workshop Tip:

Developers can **adjust** the `lambda_mult` parameter to **observe** how the variety of retrieved data changes in real-time.



Retrieval-Augmented Generation

- **Environment Setup:** Initialize LangChain and configure Groq API for high-speed LLM processing.
- **Data Preparation:** Load external documents and perform **Recursive Character Chunking** for optimal context sizing.
- **Vector Indexing:** Generate text embeddings and construct a local knowledge base using **Vector Store**.
- **Retriever Optimization:** Compare **Top-K Similarity Search** vs. **Maximum Marginal Relevance (MMR)** to reduce redundancy.
- **Prompt Engineering:** Design custom templates to enforce "Fact-based" answering and prevent hallucinations.
- **Application Deployment:** Build and launch a fully functional, interactive Chatbot UI using **Gradio**.